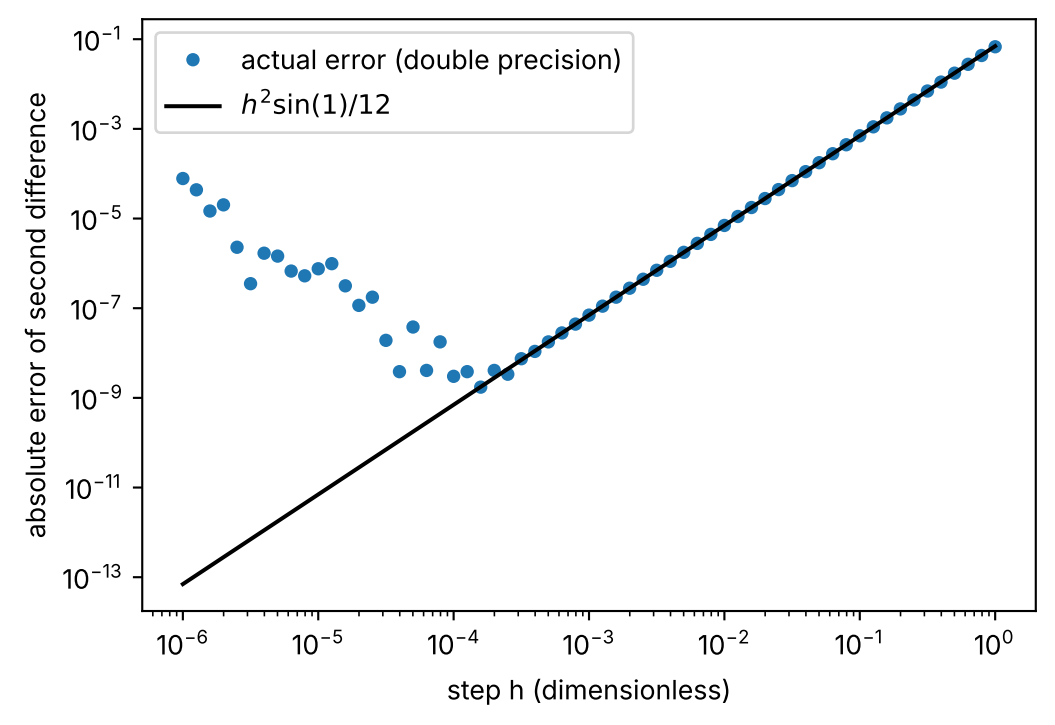
actual error (double precision)
$h^2\sin(1)/12$
absolute error of second difference
step h (dimensionless)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}